

Text Representation Research Notes

- **Sentence-BERT**

- Used to generate dense sentence-level embeddings for question-context representation.
- Relevant paper: Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.
- Reason for use: captures semantic similarity better than sparse bag-of-words methods.

- **BioBERT**

- Biomedical domain-specific BERT model.
- Relevant paper: BioBERT: a pre-trained biomedical language representation model for biomedical text mining.
- Reason for use: expected to capture biomedical terminology and clinical context better than general embeddings.

- **Hypothesis**

Dense biomedical representations should perform better than sparse TF-IDF/count-based representations, especially where medical meaning depends on context rather than isolated keywords.